

PVTRACE · REQUIREMENTS

PVTrace requirements

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DOCUMENT CONTROL

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Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/pvtrace		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-03	Amish Chadha	R3 wording: selected capacitor bank minimum of 6.45 mF decided by Amish on 2026-10-03
0.7	2026-10-02	Amish Chadha	R3 now met on paper with the capacitor bank selected at build (PVT-CAL-001 v0.5); R17 restated to USD 192, USD 27 over the target; R12 margin about 6 g
0.6	2026-10-02	Amish Chadha	R3 and R12 status notes record the 2026-10-02 decisions (PVT-DEC-001, items 3 and 4); no status changed
0.5	2026-10-01	Amish Chadha	Constructable design (PVT-DDR-003); R12 mass 1.49 kg; R17 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Status from PVT-CAL-001; decisions from PVT-DDR-001 (6.6 mF, isolation, grade thresholds); R10 and R17 wording; 33 J stored energy
0.2	2026-09-25	Amish Chadha	

First measurable requirements with
concept status for TRL 2

0.1	2026-09-25	Amish Chadha	Initial scaffold
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PVTrace requirements

These requirements are proposals for review, not yet agreed with users, and will be revised after co-design (see PVT-PRB-001). The status column gives the design's position against each target from the calculations in PVT-CAL-001 (TRL 3). Two requirements are **not met** on paper (R13 display readability, R16 out of scope), ten are met including R3 with the capacitor bank selected at build, two are at risk (R7, R8) and two cannot be verified until hardware exists (R6, R9). R17 is reported against the value-engineering target (`budget_usd` , USD 165, a hypothetical control target, not a limit; Amish, 2026-10-01): the constructable design of PVT-DDR-003 with the keypad, status LEDs and a spare capacitor is estimated at USD 192, USD 27 over the target. The design choices were decided by Amish on 2026-09-25 (go with recommendation; PVT-DDR-001 and PVT-DDR-002).

The **reference module** is a 144 half-cell module of about 450 W: open-circuit voltage (Voc) about 49.5 V, short-circuit current (Isc) about 11.6 A. The **high-current case** is a 108 half-cell 182 mm module of about 410 W (Voc about 37.5 V, Isc about 13.9 A). The **high-voltage case** is two older 60-cell 250 W modules in series (Voc about 75 V, Isc about 8.9 A).

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PVT-CAL-001)
R1	Voltage range	0 to 100 V DC open circuit, including cold-weather Voc rise; hard limit below 120 V	Component ratings; calculation	Met by design review: high-voltage case 83 V at -10 °C; 160 V capacitors, 150 V MOSFETs. Two 450 W modules in series (109 V cold) are refused
R2	Current range	0 to 20 A short circuit	Shunt and switch ratings	Met by design review: 20.48 A full scale; shunt 1.6 W of 3 W
R3	Sweep duration, to limit capacitive error in high-efficiency cells	20 to 200 ms from Isc to Voc for any module in range	Sweep model; later bench sweep	Met on paper with a selected bank (changed from not met on 2026-10-02): Amish decided to measure the capacitors at build and select a set of at least 6.4 mF (PVT-DEC-001, item 4); on 2026-10-03 he decided the minimum is 6.45 mF. The high-current case then sweeps in 20.1 ms at 6.45 mF (19.9 ms at exactly 6.4 mF); longest 81 ms. Capacitors as bought reach only 16.4 ms at -20 %
R4	Curve resolution	200 or more V-I pairs per sweep	Sample-rate calculation	Met: 410 pairs or more
R5	Measurement accuracy after calibration	Voltage and current within ± 1 % of reading ± 0.1 % of full scale	Error budget; later check against a calibrated meter	Met on paper: about ± 0.2 % of reading ± 0.03 % of full scale after calibration
R6	Repeatability	Pmax within ± 1 % over 3 consecutive sweeps at stable irradiance (± 1 % change)	Later field test	Not verifiable at TRL 3; instrument noise about 0.08 % per sample
R7	Irradiance and temperature	Plane-of-array irradiance within ± 5 % from a reference cell; module back temperature within ± 2 °C	Calibration plan; CalRig for the temperature probe	At risk: needs a reference cell calibrated within ± 4.6 %
R8	STC translation	Pmax translated to STC per IEC 60891 with an expanded uncertainty of ± 5 % or better at 700 W/m ² or more	Uncertainty budget	At risk: ± 3.7 to ± 5.4 %; met if the reference cell is within ± 4.5 %
R9	Fault flags	Flag steps from bypass diode conduction (loss of one substring or more), raised series resistance, lowered shunt resistance and an Isc deficit against irradiance; shown within 5 s of the sweep	Rule definitions; later tests on modules with known faults	Not verifiable at TRL 3; about 7 or more samples per volt resolve a substring step
R10	Second-life grading	Grade A, B, C or reject from STC Pmax against nameplate, with	Rule review with partner	Met by design review; thresholds decided by Amish (PVT-DDR-001 item 8); a partner can still tune the table

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PVT-CAL-001)
		thresholds held in an editable table that a partner can change		
R11	Safe connection and disconnection	Reverse polarity and over-voltage detected before the load connects; the circuit is never opened above 0.5 A; load capacitors below 30 V within 2 s after each sweep and below 60 V within 60 s if the controller fails	Circuit review; discharge calculation	Met: 0.21 s to 30 V; 40 s passive to 60 V; load switch held up to 6 ms past 99 % of Voc so it opens below 0.5 A
R12	Handheld	Mass 1.5 kg or less including leads and sensor pod; enclosure 250 x 150 x 100 mm or less	Mass estimate; model	Met: 1.49 kg for the constructable design (PVT-DDR-003), about 6 g to spare after the keypad and LEDs (was 14 g), accepted by Amish on 2026-10-02 with the prototype weighed at TRL 4 (PVT-DEC-001, item 3); 234 x 144 x 80 mm over bumpers, 98 mm to the knob and hood tops
R13	Field conditions	Operate at 0 to 45 °C ambient in full sun; IP54 when closed; display readable in direct sun	Datasheets; heat model; later field check	IP54 met by design; display readability not met on paper: sun hood fitted (PVT-DDR-002), readability to be shown in a field check (TRL 4); heat at risk (about 56 °C inside a light case at 45 °C in sun)
R14	Battery life	8 h of field use or 200 sweeps per charge; USB-C charging	Power budget	Met: 10.3 h
R15	Open data	Every sweep saved as CSV (raw V-I pairs, irradiance, temperature, time, module ID, flags, grade); export by USB or Wi-Fi to a phone	Design review (no firmware at TRL 3)	Met by design review
R16	Second-life safety screening	Detect insulation faults (cracked backsheet, wet leakage) before a module is resold	Not in this instrument	Not met: out of scope; an insulation tester is needed alongside PVTrace
R17	Cost	Parts for one prototype against the value-engineering target in <code>budget_usd</code> (USD 165)	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Value-engineering target: USD 165. Estimated cost of the constructable design: USD 192 (USD 27 over the target)

■ Assumptions

- Module curves are modeled with a single-diode model fitted to datasheet Voc, Isc, Vmp and Imp; sweep times are the time for a 6.6 mF load (± 20 %) to reach 99 % of Voc, with 56.6 m Ω loop resistance (PVT-CAL-001).
- The ADC delivers at least 25,000 V-I pairs per second in practice (half the rated rate of a 100 kS/s two-channel converter, to allow for SPI and firmware overhead).

- The reference cell is calibrated once against a pyranometer or a module of known power under clear sky; its uncertainty (about ± 3 to 5 %) dominates R8.
- The 20 ms lower bound in R3 is a working figure for high-efficiency (PERC, TOPCon, HJT) cells, whose capacitance distorts fast sweeps; it could not be checked against literature at TRL 3 and stays a working figure. PVT-CAL-001 shows the capacitive current error is about $C_{\text{module}} / C_{\text{load}}$, so it depends on the load capacitance more than on the sweep time.
- 700 W/m² follows IEC 61829 guidance for converting to STC ([Seaward](#)).

Safety: The tracer connects directly to energized PV modules at up to 100 V DC and 20 A, and stores up to 33 J in its load capacitors. R11 exists to keep the user safe; it must not be relaxed to meet cost.